Classical training of conditional instantaneous quantum polynomial-time Born machines

Oriol Balló-Gimbernat,[1,2,3,*] Emiliano Tolotti,[4] Laia Domingo (Superpositions o CVC?),[2] Adan Garriga,[1] and Fernando Vilariño[2,3]

[1] *Eurecat, Centre Tecnològic de Catalunya, Barcelona, Spain*
[2] *Centre de Visió per Computador (CVC), Barcelona, Spain*
[3] *Universitat Autònoma de Barcelona (UAB), Barcelona, Spain*
[4] *Department of Information Engineering and Computer Science, University of Trento, Via Sommarive 9, 38123 Trento, Italy*

(Dated: October 6, 2026)

Instantaneous quantum polynomial-time (IQP) Born machines offer a scalable approach to quantum generative modeling by decoupling model optimization from quantum hardware. However, their standard fixed-input formulation models a single distribution. In this work, we introduce a conditional framework that encodes conditions into input states before one shared learned IQP unitary. The input preparation need not have strict IQP form, provided its reference state supports efficient classical sampling and amplitude-ratio evaluation. These operations enable unbiased classical estimation of Pauli-Z moments and their gradients, preserving classical trainability. We empirically validate this approach on a sixteen-qubit discrete Gaussian family and the Curie–Weiss model. We also demonstrate the computational feasibility of classical conditional training from 64 to 2048 qubits. Furthermore, we show how input-state encoding can impose exact output symmetries in both conditional and unconditional IQP Born machines. More broadly, this framework extends classically trainable IQP Born machines to conditional generation.

I. INTRODUCTION

Conditional generative models approximate a family of target distributions $p(\cdot \mid c)$ with a model $q_{\boldsymbol{\theta}}(\cdot \mid c)$ whose learned parameters are shared across conditions. Classical examples include conditional generative adversarial networks [1–3], conditional variational autoencoders [4–6] and conditional diffusion models [7–9]. In the quantum domain, quantum circuit Born machines (QCBMs) [10, 11] have also been adapted to conditional tasks [12]. Conditioning can be introduced through input-state preparation or dedicated condition registers before a shared parameterized circuit [13–16]. Alternatively, conditional samples can also be obtained from a learned joint Born distribution through postselection [17]. Beyond QCBMs, quantum diffusion models have explored conditional generation of both classical data and quantum states [18–20].

A particularly promising architecture in this space is the instantaneous quantum polynomial-time (IQP) Born machine. Sampling from certain IQP output families is conjectured to be classically intractable under standard complexity-theoretic assumptions [21–25]. Yet, for training via the maximum mean discrepancy (MMD) objective [26], which is common for QCBMs [10, 27], the required Pauli-Z expectation values and parameter gradients admit efficient classical estimators [28]. These estimators build on Van den Nest's techniques for computationally tractable states [29]. However, the standard formulation models a single target distribution per circuit. Extending IQP Born machines to conditional generation requires a conditioning mechanism that preserves classical access to the training observables.

In this work, we develop a classically trainable framework for conditional IQP Born machines. A condition-dependent input state $|\phi(c)\rangle$ is followed by one shared learned unitary $U_{\boldsymbol{\theta}}$, giving $q_{\boldsymbol{\theta}}(x \mid c) = |\langle x|U_{\boldsymbol{\theta}}|\phi(c)\rangle|^2$. We establish sufficient conditions on the encoder for efficient classical estimation of the conditional training observables and their gradients, and construct encoders satisfying these conditions. Specifically, the reference amplitudes $\psi_c(z) = \langle z|H^{\otimes n}|\phi(c)\rangle$ must support efficient sampling and amplitude-ratio evaluation. Building on sparse-observable methods [29, 30] and IQP training methods [28], we derive explicit unbiased estimators of conditional Pauli-Z moments and phase derivatives, with second moments bounded by one and four, respectively. Each specified quantity therefore admits additive error ϵ at fixed confidence using $O(\epsilon^{-2})$ reference samples. We combine these estimators with an unbiased finite-sample conditional MMD objective and its gradients, enabling classical optimization of a single IQP Born machine across conditions. Figure 1 summarizes the framework.

We evaluate this framework across several tasks. First, using a 16-qubit discrete Gaussian family, we demonstrate that a shared model successfully maps input conditions to translated output profiles. Next, we analyze temperature-dependent changes in distributional shape using the Curie–Weiss model. Here, we compare a single conditional IQP model, evaluated at three held-out conditions, against independent circuits trained directly at each. Although the independent baselines require three times the parameters and optimization steps, our shared model achieves a lower mean full-distribution total vari-

* oriol.ballo@eurecat.org

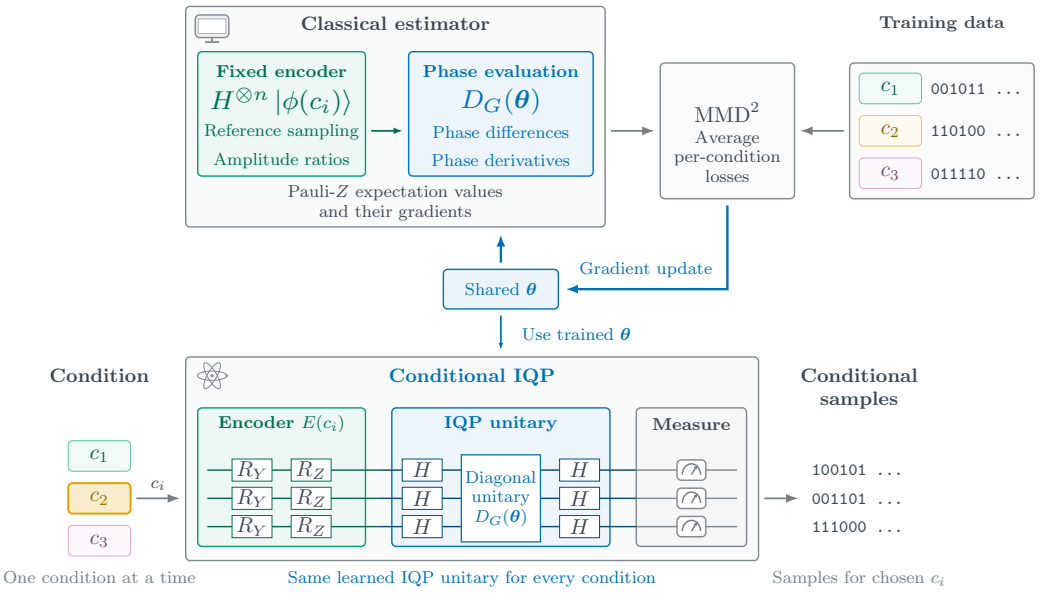


FIG. 1. Proposed framework for classical training and quantum sampling of conditional instantaneous quantum polynomial-time (IQP) Born machines. An encoder maps each condition c_i to an input state $|\phi(c_i)\rangle$. During classical training, samples and amplitude ratios from the reference state $H^{\otimes n}|\phi(c_i)\rangle$, together with phase differences induced by the IQP layer, yield estimates of conditional Pauli-Z moments and their gradients. These estimates enter the squared-MMD objective, averaged across training conditions to optimize one shared parameter vector $\boldsymbol{\theta}$. During quantum sampling, a chosen condition c_i determines the input state. Applying the trained IQP unitary $U_{\boldsymbol{\theta}}$ and measuring in the computational basis produces samples from the learned conditional distribution $q_{\boldsymbol{\theta}}(\cdot \mid c_i)$.

ation distance (TVD) at $\beta = 1.0$ and at the extrapolation point $\beta = 1.6$. The independent models yield lower mean errors at $\beta = 0.4$, where the uniform encoder-only baseline has the lowest mean error. We further assess scalability from 64 to 2048 qubits by tracking the MMD^2 at held-out temperatures throughout optimization. Furthermore, we show how input-state encoding can enforce exact output symmetries in both conditional and unconditional IQP Born machines. For the Curie–Weiss model, we compare product-state and spin-flip invariant temperature encoders under matched optimization budgets to assess how this constraint affects distributional fit. A fixed-input example illustrates the same symmetry mechanism in an unconditional model.

The remainder of the paper is organized as follows. Section II introduces the conditional IQP model and its classical estimators. Section III develops the conditional MMD training protocol. Section IV describes the numerical experiments and reports their results. Section V shows how input-state preparation can enforce output symmetries. Finally, Sections VI and VII discuss the implications and limitations of our findings and summarize the contributions.

II. CONDITIONAL IQP MODEL

The proposed model separates the dependence on the condition from the learned unitary transformation. An encoder maps each $c \in \mathcal{C}$ to an input state $|\phi(c)\rangle$, while one parameterized IQP unitary $U_{\boldsymbol{\theta}}$ acts at every condition. For a target family $\{p(\cdot \mid c) : c \in \mathcal{C}\}$ over n-bit strings, the model distribution is

$$q_{\boldsymbol{\theta}}(x \mid c) = \left|\langle x| U_{\boldsymbol{\theta}} |\phi(c)\rangle\right|^2, \qquad x \in \{0,1\}^n. \tag{1}$$

The encoder determines how the generative model responds to c; the parameterized ansatz determines the transformation shared by the family. Throughout this work, "IQP unitary" refers only to the learned block $U_{\boldsymbol{\theta}}$. The state-preparation circuit need not have IQP form, so the complete conditional circuit need not belong to the strict IQP circuit class. The central question is which condition-dependent input states preserve classical access to the observables and gradients required for training. The choice of encoder also restricts the conditional functions the model can represent. For rotation-based data encoding, the encoding generators determine accessible Fourier frequencies, while the learned circuit controls their coefficients [31]. Multidimensional Fourier analyses further characterize the dependence on encoding and circuit architecture [32]. These expressivity results motivate treating encoder design separately from the classical estimation criterion.

We parameterize the shared IQP unitary by an undirected graph $G = (V, E)$ with $n = |V|$ vertices. Each vertex $i \in V$ contributes a one-body generator Z_i, and each edge $\{i, j\} \in E$ contributes a two-body generator $Z_i Z_j$. The unitary is

$$U_{\boldsymbol{\theta}} = H^{\otimes n} D_G(\boldsymbol{\theta}) H^{\otimes n}, \tag{2}$$

where

$$D_G(\boldsymbol{\theta}) = \exp\left(i \sum_{i \in V} \theta_i Z_i + i \sum_{\{i,j\} \in E} \theta_{ij} Z_i Z_j \right). \tag{3}$$

Because all Pauli-Z generators commute, the corresponding phase gates constitute a commuting diagonal layer. While the concrete implementation evaluated in this work employs at most two-body generators, this architecture naturally generalizes to arbitrary k-body interactions.

Classical training does not require samples from the circuit's output distribution. It instead uses the reference state immediately before the diagonal IQP layer. Define

$$\psi_c(z) = \langle z | H^{\otimes n} | \phi(c) \rangle, \tag{4}$$

so that $|\psi_c(z)|^2$ is the computational-basis distribution of the reference state associated with condition c. We restrict attention to encoders for which, over the relevant conditions c and bit masks a, one can efficiently

1. sample $z \sim |\psi_c(z)|^2$, and
2. evaluate the amplitude ratio

$$R_c(z, a) = \frac{\psi_c(z \oplus a)}{\psi_c(z)} \tag{5}$$

for sampled z.

We refer to these two operations as the sampling-and-ratio interface.

These operations expose the two quantities changed by a Pauli-X mask. Sampling selects a reference-basis string z, while the ratio compares its amplitude with that of the paired string $z \oplus a$. The diagonal IQP layer contributes the phase difference between the same two strings. Their product gives one Monte Carlo contribution to a Pauli-Z moment [29].

The ratio is required only on the support of $|\psi_c|^2$, since strings with $\psi_c(z) = 0$ are never sampled. Computationally tractable states in the sense of Van den Nest [29], which admit efficient computational-basis sampling and amplitude evaluation, provide a sufficient realization of this interface under suitable numerical precision assumptions [29].

For a product-state encoder $|\phi(c)\rangle = \bigotimes_{i=1}^n |\phi_i(c)\rangle$, the reference amplitudes factorize as $\psi_c(z) = \prod_i \psi_{c,i}(z_i)$, where $\psi_{c,i}(b) = \langle b | H | \phi_i(c) \rangle$. Reference samples can therefore be generated by drawing each bit independently, and the amplitude ratio is

$$R_c(z, a) = \prod_{i : a_i = 1} \frac{\psi_{c,i}(1 - z_i)}{\psi_{c,i}(z_i)}. \tag{6}$$

When the local amplitudes have constant arithmetic evaluation cost, both operations require at most $O(n)$ local operations.

Efficient classical estimation of the observables used for training does not imply efficient classical sampling from $q_{\boldsymbol{\theta}}(\cdot \mid c)$. Furthermore, existing IQP sampling-hardness results apply to specified circuit families and do not establish hardness for arbitrary condition-dependent input states or particular trained parameters. Lastly, practical inference on a quantum computer requires efficient preparation circuits for both $|\phi(c)\rangle$ and the IQP unitary $U_{\boldsymbol{\theta}}$, which the classical sampling-and-ratio interface does not inherently provide.

A. Conditional Pauli estimator and gradients

The conditional MMD objective in Sec. III reduces to Pauli-Z moments. For a mask a, write $Z_a = \bigotimes_{i=1}^n Z_i^{a_i}$ and denote its model expectation by $\mu_q(a \mid c)$. We now show how reference-state access yields an estimator of this quantity and its phase derivatives.

We write the diagonal IQP layer as

$$D_{\boldsymbol{\theta}} |z\rangle = e^{i f_{\boldsymbol{\theta}}(z)} |z\rangle, \tag{7}$$

with

$$f_{\boldsymbol{\theta}}(z) = \sum_{j=1}^{P} \theta_j \chi_{g_j}(z), \qquad \chi_{g_j}(z) = (-1)^{g_j \cdot z}, \tag{8}$$

where P is the number of generators and $g_j \in \{0, 1\}^n$ specifies the support of the jth diagonal Pauli generator. The phase change induced by flipping the bits selected by a is

$$\Delta_a f_{\boldsymbol{\theta}}(z) = f_{\boldsymbol{\theta}}(z \oplus a) - f_{\boldsymbol{\theta}}(z). \tag{9}$$

Using the reference amplitudes and amplitude ratio defined in Eqs. (4) and (5), define

$$\begin{aligned} W_{c,a}(z; \boldsymbol{\theta}) &= R_c(z, a) e^{i \Delta_a f_{\boldsymbol{\theta}}(z)}, \\ Y_{c,a}(z; \boldsymbol{\theta}) &= \mathrm{Re}\, W_{c,a}(z; \boldsymbol{\theta}). \end{aligned} \tag{10}$$

The derivative of the phase difference is

$$d_{j,a}(z) = \chi_{g_j}(z \oplus a) - \chi_{g_j}(z) \in \{-2, 0, 2\}. \tag{11}$$

The following proposition specializes sparse-observable amplitude-ratio estimation [30] to conditional IQP moments and their phase derivatives. For computationally tractable reference states, efficient estimation also follows from Van den Nest's sparse-operator theorem [29].

Proposition 1 (Classical estimation from reference-state access)**.** *Fix a condition c, a mask a, and a polynomial-size diagonal Pauli phase function as in Eq. (8). Suppose that the normalized reference state is independent of $\boldsymbol{\theta}$, and that independent sampling from $|\psi_c|^2$ and evaluation of $R_c(z,a)$ on its sampled support are efficient. For $z \sim |\psi_c|^2$, with exact sampling and arithmetic, the random variables*

$$\begin{aligned} Y &= \mathrm{Re}\, W_{c,a}(z;\boldsymbol{\theta}), \\ Y'_j &= \mathrm{Re}[i\, d_{j,a}(z) W_{c,a}(z;\boldsymbol{\theta})] \end{aligned} \tag{12}$$

satisfy

$$\begin{aligned} \mathbb{E}[Y] &= \mu_q(a \mid c), & \mathbb{E}[Y^2] &\le 1, \\ \mathbb{E}[Y'_j] &= \partial_{\theta_j} \mu_q(a \mid c), & \mathbb{E}[(Y'_j)^2] &\le 4. \end{aligned} \tag{13}$$

The means of S independent samples are therefore unbiased, with mean-square errors at most $1/S$ and $4/S$, respectively. Each specified moment or phase derivative can be estimated to additive error ϵ with fixed confidence using $S = O(\epsilon^{-2})$ samples. Full support is not required.

Proof. Let $S_c = \{z : \psi_c(z) \neq 0\}$. Conjugating the measured observable through the final Hadamard layer gives $X_a = \bigotimes_{i=1}^n X_i^{a_i}$, so

$$\begin{aligned} \mu_q(a \mid c) &= \langle \psi_c |\, D_{\boldsymbol{\theta}}^\dagger X_a D_{\boldsymbol{\theta}}\, | \psi_c \rangle \\ &= \mathrm{Re} \sum_{z \in S_c} \psi_c(z)^* \psi_c(z \oplus a) e^{i \Delta_a f_{\boldsymbol{\theta}}(z)} \\ &= \mathbb{E}_{z \sim |\psi_c|^2}[Y]. \end{aligned} \tag{14}$$

Because the reference state is independent of $\boldsymbol{\theta}$, differentiating the finite sum gives

$$\begin{aligned} \partial_{\theta_j} \mu_q(a \mid c) &= \mathbb{E}_{z \sim |\psi_c|^2}[\mathrm{Re}\,(i\, d_{j,a}(z) W_{c,a}(z;\boldsymbol{\theta}))] \\ &= \mathbb{E}[Y'_j]. \end{aligned} \tag{15}$$

Although individual amplitude ratios can be large, normalization bounds their second moment:

$$\mathbb{E}[|W_{c,a}|^2] = \sum_{z \in S_c} |\psi_c(z \oplus a)|^2 \le 1. \tag{16}$$

Thus $\mathbb{E}[Y^2] \le 1$. Since $|d_{j,a}(z)| \le 2$, we also obtain $\mathbb{E}[(Y'_j)^2] \le 4$. The means of S independent samples consequently have mean-square errors at most $1/S$ and $4/S$. Chebyshev's inequality gives the stated fixed-confidence sample complexity. Full support is unnecessary because ratios are evaluated only on the sampled support. □

In practice, we estimate the moment using independent reference samples $z_s \sim |\psi_c|^2$,

$$\widehat{\mu}_q(a \mid c) = \frac{1}{S} \sum_{s=1}^{S} Y_{c,a}(z_s;\boldsymbol{\theta}). \tag{17}$$

Averaging the corresponding Y'_j values over the same samples estimates the phase derivative. For the standard IQP input, the reference amplitudes are uniform and $R_c(z,a) = 1$, so Eq. (14) reduces to the usual cosine estimator [29, 33].

For a circuit with r ancillary qubits discarded at measurement, the same construction uses the joint reference state and replaces each measured mask a by $(a, 0^r)$.

III. CONDITIONAL IQP TRAINING

This section formalizes the training of conditional IQP Born machines. Throughout this work, classically trainable means the training objective and its gradients can be estimated classically to additive accuracy.

A. Condition-balanced MMD objective

We optimize the shared parameter vector $\boldsymbol{\theta}$ by minimizing the squared maximum mean discrepancy (MMD) across the training conditions. For distributions p and q on $\{0,1\}^n$ and a positive-definite kernel k, the population MMD is [26],

$$\begin{aligned} \mathrm{MMD}^2(p,q) = \mathbb{E}_{x,x' \sim p}[k(x,x')] - 2\mathbb{E}_{x \sim p,\, y \sim q}[k(x,y)] \\ + \mathbb{E}_{y,y' \sim q}[k(y,y')], \end{aligned} \tag{18}$$

where paired samples are drawn independently.

Throughout this work, we use the Gaussian (Hamming) kernel

$$k_\sigma(x,y) = \exp\left[-\frac{d_H(x,y)}{2\sigma^2}\right], \tag{19}$$

where d_H is the Hamming distance and $\sigma > 0$ is the kernel bandwidth.

Let $\mathcal{C}_{\mathrm{train}}$ denote the set of training conditions and τ the set of kernel bandwidths. We optimize the model with the condition-balanced objective

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{C}_{\mathrm{train}}|} \sum_{c \in \mathcal{C}_{\mathrm{train}}} \frac{1}{|\tau|} \sum_{\sigma \in \tau} \mathrm{MMD}^2_\sigma(p(\cdot \mid c), q_{\boldsymbol{\theta}}(\cdot \mid c))\,. \tag{20}$$

The objective weights all training conditions and bandwidths equally. At each optimizer update, we independently sample one condition and one bandwidth uniformly, giving a stochastic estimate of this average.

B. Character expansion

For binary strings, the Gaussian kernel admits an exact expansion in binary characters. Let $a \in \{0,1\}^n$ be a random visible-bit mask with independent coordinates

$$a_i \sim \mathrm{Bernoulli}(p_\sigma), \qquad p_\sigma = \frac{1 - e^{-1/(2\sigma^2)}}{2}, \tag{21}$$

and define the binary character

$$\chi_a(x) = (-1)^{a \cdot x}. \tag{22}$$

The kernel can then be written as

$$k_\sigma(x,y) = \mathbb{E}_a[\chi_a(x)\chi_a(y)]\,. \tag{23}$$

For a condition c, define the target and model character moments

$$\begin{aligned}\mu_p(a \mid c) &= \mathbb{E}_{x\sim p(\cdot|c)}[\chi_a(x)],\\ \mu_q(a \mid c) &= \mathbb{E}_{x\sim q_{\boldsymbol{\theta}}(\cdot|c)}[\chi_a(x)].\end{aligned} \tag{24}$$

Substituting Eq. (23) into Eq. (18) gives

$$\mathrm{MMD}^2_\sigma(p(\cdot \mid c), q_{\boldsymbol{\theta}}(\cdot \mid c)) = \mathbb{E}_a\Big[(\mu_p(a \mid c) - \mu_q(a \mid c))^2\Big]\,. \tag{25}$$

For a Born machine, $\chi_a(x)$ is the eigenvalue of the visible Pauli operator

$$Z_a = \bigotimes_{i=1}^{n} Z_i^{a_i}, \tag{26}$$

and therefore

$$\mu_q(a \mid c) = \langle Z_a\rangle_{q_{\boldsymbol{\theta}}(\cdot|c)}. \tag{27}$$

The conditional MMD can thus be expressed entirely in terms of visible Pauli-Z moments. This representation allows classical training to leverage the IQP expectation-value estimator detailed in Sec. II A.

The bandwidth σ determines which correlation orders are weighted most strongly. Since the mask coordinates are sampled independently, its Hamming weight $K = |a|$ follows

$$\begin{aligned}K &\sim \mathrm{Binomial}(n, p_\sigma),\\ \mathbb{E}[K] &= np_\sigma,\\ \mathrm{Var}(K) &= np_\sigma(1-p_\sigma).\end{aligned} \tag{28}$$

Thus, each bandwidth induces a distribution over Pauli-string weights rather than selecting a single correlation order. Large bandwidths favor low-weight observables, whereas decreasing σ shifts weight toward higher-order correlations, with $\mathbb{E}[K] \to n/2$ as $\sigma \to 0$.

The choice of bandwidth has large implications, as it dictates not only which data correlations are weighted by the loss, but also the overall trainability of the model. Previous results demonstrate that cost functions consisting of global terms can induce barren plateaus [34, 35], even in shallow quantum circuits [36]. This creates a fundamental trade-off. Ensuring trainability may require restricting the objective to low-body correlations, which in turn limits the objective's ability to distinguish distributions that differ in higher-order correlations [37]. While various strategies have been proposed to navigate these optimization issues—including tailored generator sets, restricted initializations [38, 39], and adversarial kernel adaptation [40]—this remains an open question outside the scope of this work.

C. Unbiased finite-sample MMD and gradients

For a fixed condition c and visible mask a, let

$$T_b = \chi_a(x_b), \qquad x_b \sim p(\cdot \mid c), \tag{29}$$

for $B \geq 2$ independent target samples, and let

$$Y_s = Y_{c,a}(z_s;\boldsymbol{\theta}), \qquad z_s \sim |\psi_c|^2, \tag{30}$$

for $S \geq 2$ independent reference samples drawn independently of the target samples, with $Y_{c,a}$ defined in Eq. (10). Define the corresponding sample means

$$\overline{T} = \frac{1}{B}\sum_{b=1}^{B} T_b, \qquad \overline{Y} = \frac{1}{S}\sum_{s=1}^{S} Y_s. \tag{31}$$

A direct estimate of the squared moment difference, $(\overline{Y}-\overline{T})^2$, is biased because squaring a finite-sample mean introduces its sampling variance. We remove this bias by forming products only between independent sample pairs,

$$\begin{aligned}\widehat{\Delta^2_c}(a) = &\frac{1}{S(S-1)}\sum_{s\neq t} Y_sY_t - 2\overline{Y}\,\overline{T}\\ &+ \frac{1}{B(B-1)}\sum_{b\neq b'} T_bT_{b'}.\end{aligned} \tag{32}$$

Independence of the paired samples gives

$$\mathbb{E}\Big[\widehat{\Delta^2_c}(a)\Big] = (\mu_q(a \mid c) - \mu_p(a \mid c))^2\,. \tag{33}$$

Thus, Eq. (32) provides an unbiased finite-sample estimator of the squared character-moment difference appearing in Eq. (25).

We do not need to evaluate the pair sums explicitly. They can be written as

$$\sum_{s\neq t} Y_sY_t = \left(\sum_s Y_s\right)^2 - \sum_s Y_s^2, \tag{34}$$

and, since $T_b^2 = 1$,

$$\sum_{b\neq b'} T_bT_{b'} = \left(\sum_b T_b\right)^2 - B. \tag{35}$$

We can therefore evaluate the estimator in linear rather than quadratic time in the numbers of target and reference samples.

The same construction gives an unbiased estimator of the phase gradients. Let

$$Y'_s = \partial_{\theta_j} Y_{c,a}(z_s;\boldsymbol{\theta}), \tag{36}$$

with Y'_s evaluated using Eq. (15). Differentiating Eq. (32) gives

$$\partial_{\theta_j}\widehat{\Delta^2_c}(a) = \frac{2}{S(S-1)}\sum_{s\neq t} Y_sY'_t - 2\overline{T}\,\overline{Y'}, \tag{37}$$

where

$$\overline{Y'} = \frac{1}{S}\sum_s Y'_s. \tag{38}$$

Taking the expectation yields

$$\mathbb{E}\left[\partial_{\theta_j}\widehat{\Delta_c^2}(a)\right] = 2\left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)\partial_{\theta_j}\mu_q(a \mid c), \tag{39}$$

which is the gradient of the corresponding squared moment difference. At a fixed parameter vector, these population unbiasedness statements average over independent target and reference samples. During training, target minibatches are sampled with replacement from fixed training pools; the stochastic gradient is therefore unbiased for the corresponding empirical objective. An unbiased finite-sample estimate of MMD^2 may be negative.

Excluding the diagonal terms $s = t$ is important because evaluating Y_s and Y'_s from the same reference draw correlates them. Since the encoder is fixed, the reference sampling law does not depend on $\boldsymbol{\theta}$; consequently, holding the sampled reference strings fixed during automatic differentiation evaluates the required phase gradients without differentiating through the sampler.

D. Stochastic mask estimator

Equation (25) expresses the MMD as an expectation over binary masks a. Since the number of possible masks grows exponentially with n, we estimate this expectation by drawing a finite sample of masks from the distribution in Eq. (21).

The identity mask $a = 0$ satisfies

$$\mu_p(0 \mid c) = \mu_q(0 \mid c) = 1 \tag{40}$$

for every condition and therefore contributes identically zero to the MMD. We consequently sample only nonidentity masks from the conditional distribution

$$P_\sigma(a \mid a \neq 0). \tag{41}$$

The total probability assigned to nonidentity masks under the original kernel distribution is

$$Q_\sigma^+ = 1 - (1 - p_\sigma)^n. \tag{42}$$

Multiplying the conditional expectation by Q_σ^+ therefore preserves the original Hamming-RBF objective.

For M sampled masks $a_1, \ldots, a_M$, the finite-mask estimator at condition c is

$$\widehat{\mathrm{MMD}}_\sigma^2(c) = Q_\sigma^+ \frac{1}{M}\sum_{m=1}^{M}\widehat{\Delta_c^2}(a_m), \tag{43}$$

where $\widehat{\Delta_c^2}$ is the unbiased finite-sample estimator in Eq. (32). Averaging also over the mask draws recovers the full kernel MMD.

We need not evaluate repeated masks more than once. If $\mathcal{U}$ denotes the set of distinct sampled masks and m_u the multiplicity of mask u, Eq. (43) can be written equivalently as

$$\widehat{\mathrm{MMD}}_\sigma^2(c) = Q_\sigma^+ \sum_{u\in\mathcal{U}} \frac{m_u}{M}\widehat{\Delta_c^2}(u). \tag{44}$$

Combining duplicate masks, therefore, reduces the number of Pauli moments we must evaluate without changing either the sampled objective or its gradient.

IV. NUMERICAL EXPERIMENTS

We evaluate the proposed architecture on a translated Gaussian family and the Curie–Weiss model, with a supplementary scaling demonstration up to 2048 qubits. For the two sixteen-qubit benchmarks, the primary accuracy metric is the exact full-distribution total variation distance,

$$\mathrm{TVD}(p, q) = \frac{1}{2}\sum_{x\in\{0,1\}^n} |p(x) - q(x)|. \tag{45}$$

We also report the exact Hamming-profile TVD as a diagnostic,

$$\mathrm{TVD}_K(p, q) = \frac{1}{2}\sum_{k=0}^{n} |\rho_p(k) - \rho_q(k)|, \tag{46}$$

where $\rho(k)$ is the probability of observing a bitstring of weight k. Aggregation gives $\mathrm{TVD}_K(p, q) \leq \mathrm{TVD}(p, q)$. The targets are uniform within each Hamming-weight shell, but the learned models need not be, so accurate profiles can hide distributional errors. Both metrics use the same exact circuit probabilities. Training minimizes the squared-MMD objective. We select hyperparameters and checkpoints using profile TVD and evaluate the selected models using exact full-distribution TVD.

A. Discrete Gaussian translation

Our first experiment illustrates the conditioning mechanism with a moving Gaussian profile. For $n = 16$ bits, let $K = |x|$ be the Hamming weight. The target distribution translates the nominal center of a discrete Gaussian based on a condition $c \in [0.0, 1.0]$, with all bitstrings of equal weight having equal target probability.

We use a fixed product state encoder that applies the same two rotations to every qubit:

$$|\phi(c)\rangle = \bigotimes_{i=1}^{16} R_Z\left(c - \frac{1}{2}\right) R_Y(1.5)\,|0\rangle_i\,. \tag{47}$$

Measurement immediately after this encoder yields a fixed distribution; changing c modifies the input phases but leaves the outcome probabilities unchanged.

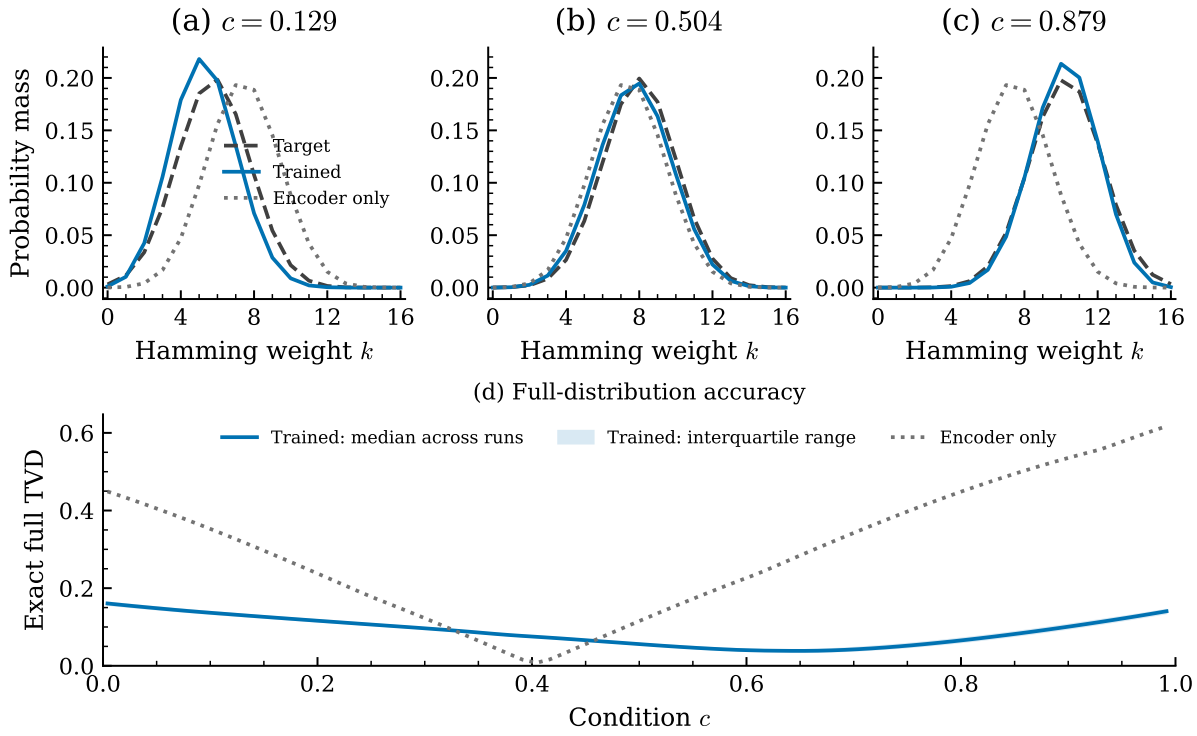


FIG. 2. Conditional Gaussian generation at sixteen qubits. Upper panels show target, trained-model and encoder-only Hamming profiles at three held-out conditions, using one run selected independently of held-out performance. The lower panel reports exact full-distribution TVD across all 80 held-out conditions. The trained curve and band show the pointwise median and interquartile range over ten independent runs; the encoder-only error is deterministic. These pointwise summaries differ from the aggregate in Table I, which takes the median of each run's mean error.

The shared unitary uses a 4×4 lattice with 16 one-body gates and 24 horizontal and vertical nearest-neighbor gates, giving 40 trainable parameters.

We train at $c \in \{0.0, 0.25, 0.50, 0.75, 1.0\}$ and evaluate across 80 held-out conditions. The median across ten training runs of the mean evaluation full TVD is 0.0876, compared with 0.2922 for the encoder alone (Table I). The corresponding trained profile TVD is 0.0670, indicating some additional error within Hamming shells. Figure 2 shows the translated profiles and full-TVD dependence on the condition. Appendix A 2 gives the training and evaluation protocol.

TABLE I. Exact Gaussian full-distribution and secondary profile TVD. Each value is the median across 10 runs of the mean error over 80 held-out conditions. Selection uses development-profile TVD.

Model	Full TVD ↓	Profile TVD ↓
Conditional IQP	0.0876	0.0670
Encoder only	0.2922	0.2922

B. Curie-Weiss spin model

The Curie–Weiss model [41] tests the architecture's ability to capture changes in distributional shape. The spins $s_i \in \{-1, +1\}$ interact with equal strength, so the target assigns equal probability to bitstrings with the same Hamming weight. As the inverse temperature β increases, the finite-size Hamming-weight profile changes from a central peak to two symmetric modes.

We encode the inverse temperature using the fixed product phase encoder

$$|\phi(\beta)\rangle = \bigotimes_{i=1}^{16} R_Z(0.75\beta) R_Y\left(\frac{\pi}{2}\right) |0\rangle_i \,. \tag{48}$$

The encoder-only output is uniform for every β. The parameterized circuit uses all one- and two-body generators, giving 136 parameters. We train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$ and evaluate at $\beta \in \{0.4, 1.0, 1.6\}$.

We compare the shared conditional IQP across conditions with two independently tuned baseline families. Each family fits three separate circuits, one at each reporting temperature. The independent conditional IQP per condition retains the corresponding temperature encoder, while the independent unconditional IQP per condition uses the ordinary input.

We tune the shared model on development temperatures and each independent family jointly across its fitted temperatures, using exact profile TVD for selection. The full-TVD results evaluate those same selected circuits. Appendix A 3 a details hyperparameter optimization and model selection.

Table II reports the exact full-distribution TVDs. The shared conditional IQP has the lowest mean error among trained families at $\beta = 1.0$ and 1.6, with substantial variability at $\beta = 1.6$. Independent fits have lower mean errors at $\beta = 0.4$, where the uniform encoder-only law has lower mean error than every trained family. Relative to that encoder, the shared model improves mean full TVD at $\beta = 1.0$ and 1.6 but worsens it at $\beta = 0.4$. These findings describe this target family and the existing profile-based selection protocol; they do not establish a general advantage of parameter sharing.

Each independent family uses three circuits, with three times the total number of parameters and optimizer updates of the shared model. The shared model uses more total training-pool samples. These unequal aggregate resources and the different evaluation roles limit conclusions about parameter sharing alone. Appendix A 3 reports the detailed budgets.

The secondary profile diagnostic gives a different ranking. The independent models have lower mean profile TVD at $\beta = 1.0$, despite higher full TVD than the shared model. The shared model also improves the encoder's profile TVD at $\beta = 0.4$, while worsening full TVD. These differences show why profile agreement alone does not establish accurate generation of bitstrings. Appendix A 3 retains the profile results alongside the full-distribution analysis.

1. *Spin-flip invariant temperature encoding*

The zero-field Curie–Weiss target is invariant under reversal of all spins, $p(x \mid \beta) = p(\bar{x} \mid \beta)$. We impose this constraint through the input state while retaining the temperature map and all 136 independent circuit phases. Writing $t = 0.75\beta$ and $F = X^{\otimes n}$, we project the product encoder onto the $+1$ eigenspace of F:

$$|\phi_+(\beta)\rangle = \frac{(I+F)\,|\phi(\beta)\rangle}{\sqrt{2[1+\cos^n(t)]}}. \tag{49}$$

The normalization follows from $\langle\phi(\beta)|F|\phi(\beta)\rangle = \cos^n(t)$. Every IQP generator commutes with F, so $FU_{\boldsymbol{\theta}}\,|\phi_+(\beta)\rangle = U_{\boldsymbol{\theta}}\,|\phi_+(\beta)\rangle$ and the output satisfies $q_{\boldsymbol{\theta},+}(x \mid \beta) = q_{\boldsymbol{\theta},+}(\bar{x} \mid \beta)$ for every parameter vector. This conditional input applies the commuting-symmetry construction of Recio-Armengol *et al.* [28, Sec. 6], which we discuss in Sec. V. Independent learned phases can still break permutation symmetry, so spin-flip invariance does not imply uniform probabilities within Hamming shells.

The projected encoder preserves efficient classical access to the reference state. For $k = |z|$,

$$\psi_{+,\beta}(z) = \frac{\mathbf{1}_{k\ \text{even}}[\cos(t/2)]^{n-k}[-i\sin(t/2)]^k}{\sqrt{[1+\cos^n(t)]/2}}. \tag{50}$$

We sample k from a binomial law with success probability $\sin^2(t/2)$ conditioned on even parity, then choose a uniformly random weight-k bitstring. For a sampled z, the amplitude ratio is

$$R_{+,\beta}(z,a) = \begin{cases} [-i\tan(t/2)]^{|a|-2z\cdot a}, & |a| \text{ even}, \\ 0, & |a| \text{ odd}. \end{cases} \tag{51}$$

These operations require polynomial resources and enter the same conditional estimator and training objective. A deterministic preparation circuit uses a running parity wire within the existing register and $O(n)$ gates, without an ancillary qubit or postselection.

We compare the product-state encoder with the spin-flip invariant encoder using separate, matched `Adam` searches and five paired evaluation seeds. Both models use the same fitting and development temperatures and training budgets. The product-input model is fitted independently for this comparison, so its errors differ from those in Table II. Appendix A 3 c specifies the comparison protocol. We evaluate exact full-distribution TVD, secondary profile TVD, and the same-temperature spin-flip error

$$D_{\text{flip}}(q \mid \beta) = \frac{1}{2}\sum_x |q(x \mid \beta) - q(\bar{x} \mid \beta)|\,. \tag{52}$$

The symmetry guarantee holds independently of optimization; distributional fit remains an empirical question.

The invariant input lowers mean full-distribution TVD at all three reporting temperatures in this matched comparison (Table III). At $\beta = 0.4, 1.0, 1.6$, the mean errors decrease from 0.2505, 0.2192, and 0.1936 to 0.1056, 0.1645, and 0.1746, respectively. Full TVD improves in all five paired seeds at $\beta = 0.4$ and 1.0, and in four of five at $\beta = 1.6$. The profile errors are close at $\beta = 1.0$ and 1.6, so profile agreement alone would obscure much of this improvement. The invariant input's spin-flip error remains below 10^{-12} at every reporting temperature. This exact symmetry does not enforce uniform probabilities within Hamming shells, and appreciable full-distribution errors remain.

Classical averaging of the product-input output gives lower mean full TVD at $\beta = 0.4$, 0.0585 rather than 0.1056 for the invariant input, without additional training. At $\beta = 1.0$ and 1.6, the invariant input retains lower mean full TVD than the averaged output. Appendix A 3 c reports this diagnostic, secondary profile errors and paired differences. Thus, input projection improves full-distribution fit relative to the trained product-input model in this comparison, but does not give uniformly lower mean error than symmetry enforcement after sampling.

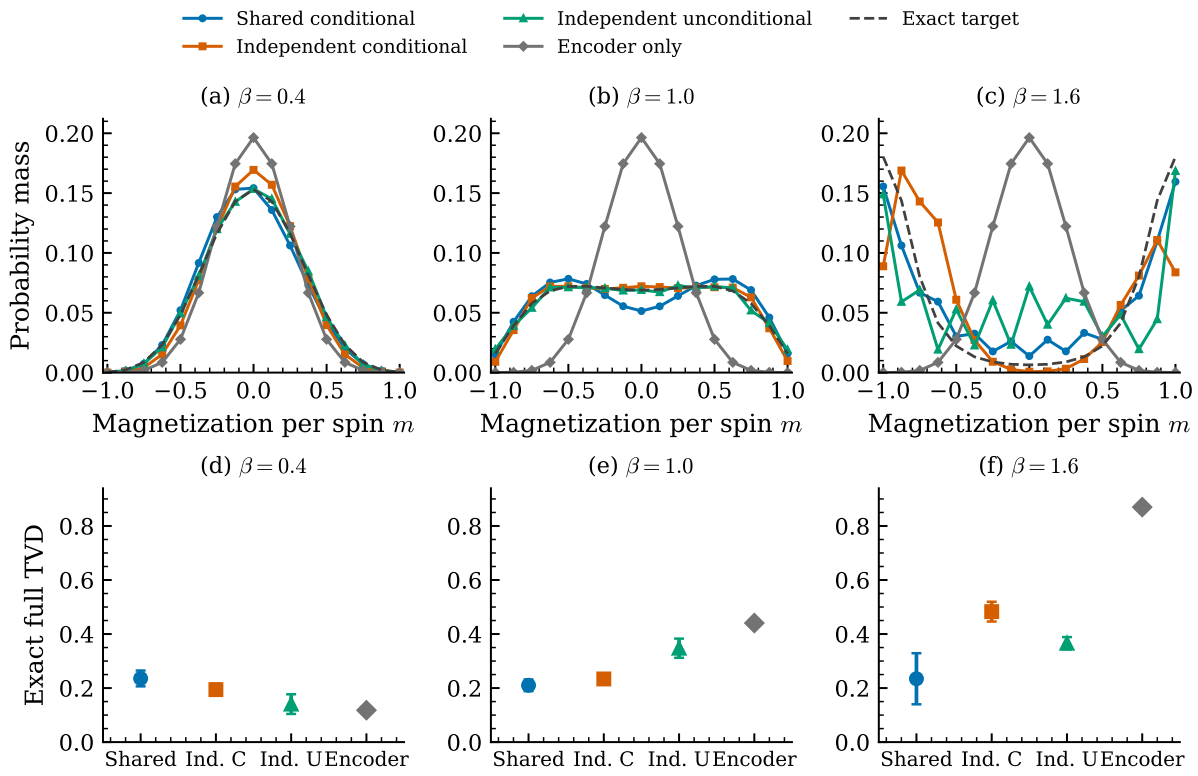


FIG. 3. Curie–Weiss profiles and full-distribution accuracy at sixteen qubits. Columns correspond to $\beta = 0.4, 1.0, 1.6$. Upper panels show mean exact raw magnetization profiles, with $m = (2k-16)/16$, and dashed exact targets. Lower panels show mean exact full-distribution TVD with sample standard deviations across five independent runs; the encoder is deterministic. Errors are averaged across individual runs, rather than computed from mean profiles. The shared model is evaluated at held-out temperatures; independent circuits are fitted directly at each reporting temperature. All trained circuits were selected using profile TVD.

TABLE II. Exact raw Curie–Weiss full-distribution TVD at sixteen qubits. Trained rows report means and sample standard deviations over 5 independent runs; the encoder is deterministic. Shared reporting temperatures are held out, while independent circuits are fitted at each reporting temperature. Bold entries mark the lowest mean error, including the encoder. Selection remains profile TVD.

Model	Evaluation	Full-distribution TVD ↓		
		$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Encoder only	Baseline	**0.1180**	0.4406	0.8699
Shared conditional IQP across conditions	Held-out	0.2356 ± 0.0293	$\mathbf{0.2102 \pm 0.0217}$	$\mathbf{0.2345 \pm 0.0945}$
Independent conditional IQP per condition	Fitted	0.1941 ± 0.0221	0.2337 ± 0.0077	0.4829 ± 0.0366
Independent unconditional IQP per condition	Fitted	0.1406 ± 0.0364	0.3475 ± 0.0355	0.3660 ± 0.0227

C. Scaling

The scaling experiment uses optimizer and evaluation settings that differ from the tuned sixteen-qubit comparison. It assesses computational feasibility; transfer of the selected sixteen-qubit configuration has not been tested.

We evaluate the computational feasibility of classical training at $n \in \{64, 128, 256, 512, 1024, 2048\}$ visible qubits, without ancillas. We retain the fixed product phase encoder $R_Z(0.75\beta)R_Y(\pi/2)$ on every qubit. The parameterized unitary contains independent one-body phases and horizontal and vertical nearest-neighbor pair phases on an open rectangular lattice. For an $L_x \times L_y$ lattice, the ansatz has $3n - L_x - L_y$ parameters, so the parameter count grows linearly with the number of qubits. The lattice dimensions and optimization hyperparameters are listed in Appendix A.

At each size, we train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$ and track sampled MMD^2 throughout optimization at held-out temperatures $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$. Appendix A 4 specifies the training and reporting protocol.

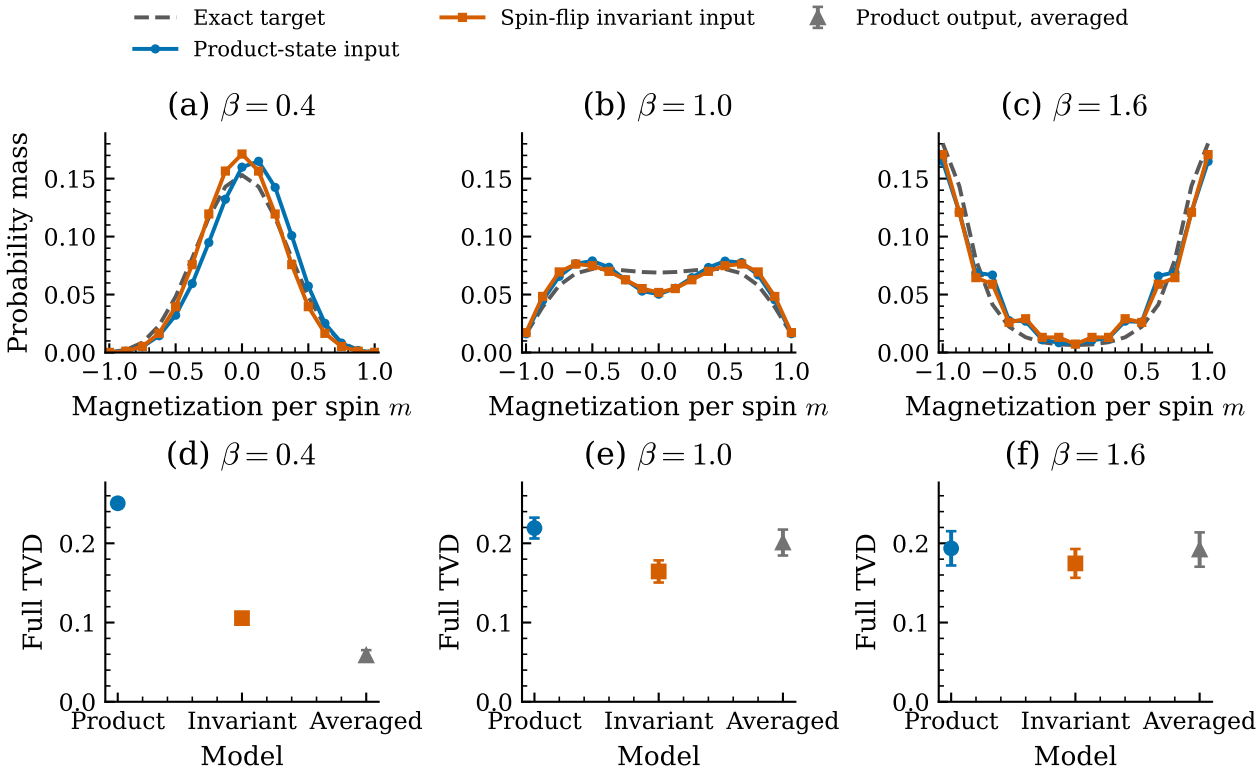


FIG. 4. Curie–Weiss profiles and full-distribution accuracy for product-state and spin-flip invariant inputs. Both models have sixteen visible qubits and all 136 independent phases. Columns correspond to $\beta = 0.4, 1.0, 1.6$. Upper panels show mean exact raw magnetization profiles over five paired final seeds and dashed exact targets, with $m = (2k - 16)/16$. Lower panels report mean exact full-distribution TVD and sample standard deviations, including classical averaging of the trained product output without retraining. Reporting temperatures enter neither fitting nor current selection but have historical exposure. Selection uses profile TVD; reported errors average individual-seed errors rather than errors of mean profiles.

TABLE III. Exact profile and full-bitstring TVD and same-temperature spin-flip error for the trained Curie–Weiss models; Full TVD is primary and Profile TVD is secondary. Means ± sample standard deviations over 5 paired seeds. Reporting temperatures enter neither fitting nor current selection but have historical exposure.

β	Input	Full TVD	Profile TVD	D_{flip}
0.4	Product-state input	0.2505 ± 0.0057	0.0852 ± 0.0270	0.4837 ± 0.0133
0.4	Spin-flip invariant input	0.1056 ± 0.0066	0.0513 ± 0.0072	0.0000 ± 0.0000
1.0	Product-state input	0.2192 ± 0.0132	0.0646 ± 0.0112	0.1586 ± 0.0184
1.0	Spin-flip invariant input	0.1645 ± 0.0140	0.0659 ± 0.0130	0.0000 ± 0.0000
1.6	Product-state input	0.1936 ± 0.0216	0.0987 ± 0.0154	0.0183 ± 0.0017
1.6	Spin-flip invariant input	0.1746 ± 0.0182	0.0962 ± 0.0143	0.0000 ± 0.0000

Figure 5 shows a decrease in held-out sampled MMD^2 at every system size, followed by a persistent residual. At 64 qubits, the median across runs decreases from 0.0504 at initialization to 0.0350 after 5000 updates; at 2048 qubits, it decreases from 0.0490 to 0.0417. These results demonstrate the feasibility of classically optimizing conditional IQP Born machines up to 2048 qubits. However, substantial discrepancies remain at high inverse temperature, and a decrease in sampled MMD^2 does not establish accurate recovery of the full target distribution. The same bodyness targets are used at every size, so the corresponding kernel bandwidths vary with system size. Absolute MMD^2 values should therefore not be interpreted as direct comparisons of distributional fidelity across sizes.

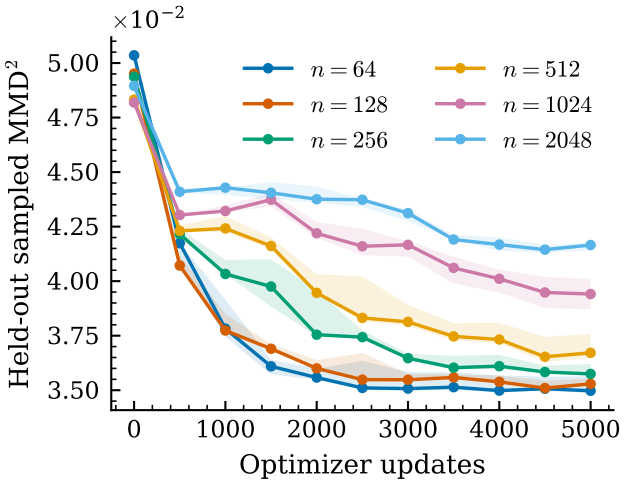


FIG. 5. Classical training of conditional IQP Born machines from 64 to 2048 qubits. Lines and shaded bands show the median and interquartile range of held-out sampled MMD^2 across three independent training runs. Evaluation conditions are $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$. The same bodyness targets $w \in \{2, 4\}$ are used at every system size, implying different kernel bandwidths.

V. INPUT-STATE INDUCED SYMMETRIES IN IQP BORN MACHINES

Encoder design determines both the dependence on a condition and the constraints satisfied by the output law. The projected Curie–Weiss input in Sec. IV B 1 combines these roles. More generally, if a bit-flip operator X_s commutes with the learned unitary and each encoded input is an eigenstate of X_s, the conditional output is invariant under that flip at every condition and parameter vector. We illustrate the same construction with a fixed unconditional input while preserving classical trainability. Such constraints provide inductive biases in variational quantum models [42].

Choose any nonzero bit mask $s \in \{0,1\}^n$ and prepare

$$|\phi_s\rangle = \frac{|0^n\rangle + |s\rangle}{\sqrt{2}}. \tag{53}$$

This state satisfies $X_s |\phi_s\rangle = |\phi_s\rangle$, where $X_s = \bigotimes_{i=1}^n X_i^{s_i}$ flips the qubits selected by s. Since the IQP unitary commutes with X_s, its output remains invariant under the same flip, even when all learned phases vary independently. Consequently,

$$q_{\boldsymbol{\theta}}(x) = q_{\boldsymbol{\theta}}(x \oplus s) \qquad \text{for every } x \text{ and } \boldsymbol{\theta}. \tag{54}$$

The reference amplitudes are uniform on the strings satisfying $s \cdot z = 0$ and vanish elsewhere, evaluating the dot product modulo two. Sampling uniformly subject to this single parity constraint is efficient. For a sampled string z, the amplitude ratio is

$$R_s(z,a) = \begin{cases} 1, & s \cdot a = 0, \\ 0, & s \cdot a = 1. \end{cases} \tag{55}$$

The encoded input therefore satisfies the sampling-and-ratio interface, preserving classical access to the moments and gradients required for training.

For example, $s = 110$ on three qubits gives $|\phi_s\rangle = (|000\rangle + |110\rangle)/\sqrt{2}$ and enforces $q_{\boldsymbol{\theta}}(x) = q_{\boldsymbol{\theta}}(x \oplus 110)$. The ordinary $|000\rangle$ input does not enforce this relation: with only $\theta_1 = \pi/6$ nonzero, it gives $q(000) = 3/4$ and $q(110) = 0$. Figure 6 compares the two inputs under the same unitary. This selected-subset construction illustrates the commuting-symmetry recipe of Recio-Armengol *et al.*, whose demonstrated example enforces global bit-flip symmetry [28].

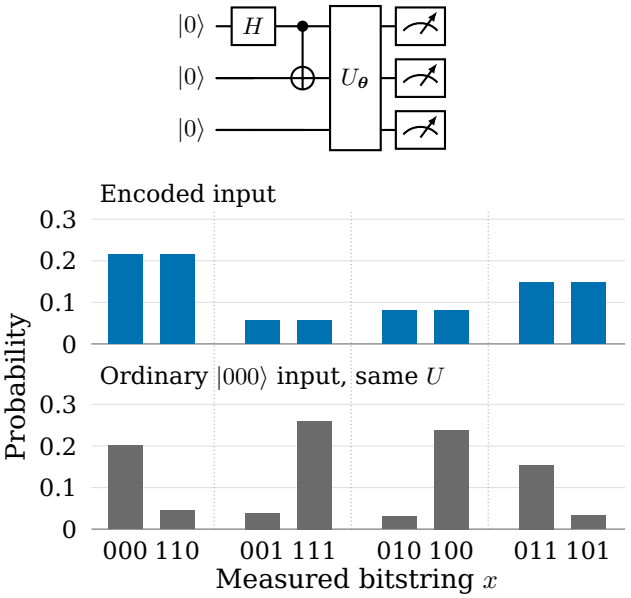


FIG. 6. The encoded input makes paired outcomes equally likely. The circuit prepares $(|000\rangle + |110\rangle)/\sqrt{2}$ before applying $U_{\boldsymbol{\theta}}$. The plots show exact probabilities at one parameter setting for the encoded input (blue) and the ordinary $|000\rangle$ input (gray), using the same unitary in both cases. Adjacent outcomes differ by 110. Their equality for the encoded input holds for every phase vector.

VI. DISCUSSION

Our numerical experiments demonstrate conditional generation with measurable residual full-distribution errors. In the Gaussian experiment, the shared model reduces aggregate full TVD relative to the encoder. In the Curie–Weiss baseline comparison, the shared model has lower mean full TVD than both independently fitted families at $\beta = 1.0$ and 1.6. At $\beta = 0.4$, the encoder-only law

has lower mean error than every trained family. Different evaluation roles and unequal aggregate training budgets prevent attributing these outcomes to parameter sharing alone.

Full-distribution TVD exposes errors that Hamming-weight aggregation can hide and changes some model rankings. Efficient estimation of the training moments therefore does not guarantee accurate recovery of the full distribution. The reported circuits were tuned and selected using profile TVD; changing the reporting metric does not optimize those circuits for full TVD. Evaluating full-TVD-based selection and hyperparameter tuning is a separate question left for future work. Herrero-Gonzalez *et al.* analyze discrepancies between classical correlator-based surrogates and deployed Born distributions [43]. Under the stated sampling and arithmetic assumptions, our estimators target the moments of the specified circuit without truncating Pauli propagation. Nevertheless, finite sampling budgets and the correlations emphasized by the kernel can leave distributional errors unresolved. Kernel-adaptive IQP training offers a possible extension beyond the fixed Gaussian kernels used here [40]; its benefit for conditional generation remains to be evaluated.

The scaling experiment demonstrates the computational feasibility of classical conditional training up to 2048 qubits, with residual sampled MMD^2 discrepancies. It uses different optimizer settings and a different ansatz from the tuned sixteen-qubit comparison, so transfer of the selected hyperparameters remains untested. The fixed bodyness targets imply size-dependent bandwidths, and the resulting MMD^2 values do not provide direct comparisons of distributional fidelity across sizes. Further study of the encoder, ansatz and optimization settings, together with diagnostics beyond sampled MMD, is needed to assess conditional accuracy at larger sizes.

Encoder design controls the conditional functions available to the model and can impose exact constraints on its output. Both the product phase encoder and its spin-flip invariant projection admit efficient state preparation, reference sampling and amplitude-ratio evaluation. The Curie–Weiss construction in Sec. IV B 1 and the unconditional illustration in Sec. V restrict the output family through the input, independently of the learned phases. In the matched Curie–Weiss comparison, projection lowers mean full TVD relative to the trained product-input model at every reporting temperature. Classical output averaging gives lower mean error at $\beta = 0.4$, whereas projection gives lower mean error at $\beta = 1.0$ and 1.6. These empirical results do not follow from the symmetry guarantee, and spin-flip invariance does not resolve all within-shell errors. More structured encoders could offer richer conditional dependence if they retain the same reference interface. Spectral Born machines extend IQP generative models through group Fourier analysis while preserving efficient classical training [44]. Investigating conditional reference-state access in that setting could broaden the framework to other forms of discrete data.

Quantum deployment requires assessing how noise and hardware constraints affect conditional accuracy. Ryu *et al.* propose logical-noise injection to decompose large IQP circuits into smaller clusters that fit qubit-limited processors [45]. This approximate strategy introduces a further potential discrepancy from the trained distribution. Its suitability for conditional models requires evaluating such errors across conditions.

Whether useful trained conditional models retain classical sampling hardness remains open. Herbst *et al.* identify average-case trainability limitations for anticoncentrating output families [46]. Bläser *et al.* propose a cross-entropy verification protocol for cubic IQP circuits with planted independent spaces [47]. Under two complexity conjectures, secret knowledge of the planted structure creates a polynomial gap between verification and classical spoofing costs, although both remain exponential. These guarantees concern a prescribed circuit family and do not establish hardness for our trained conditional models. Practical quantum advantage would require evidence of useful conditional generation together with an appropriate classical comparison and a justified sampling-hardness claim.

VII. CONCLUSIONS

We introduced a conditional formulation for IQP Born machines that combines a condition-dependent encoder with a single shared learned unitary. Efficient reference-state sampling and amplitude-ratio evaluation enable unbiased classical estimation of conditional Pauli-Z moments and phase gradients, supporting optimization without quantum circuit evaluations. The same interface also accommodates input states that impose exact output symmetries.

The Gaussian experiment demonstrates improved full-distribution accuracy relative to the encoder. The Curie–Weiss comparison shows temperature-dependent tradeoffs against independent fits, and the matched encoder comparison shows lower mean full TVD with a spin-flip invariant input than with a trained product input. Classical output averaging remains a stronger comparator at one reporting temperature. These results evaluate circuits selected using profile TVD. Supplementary runs establish the computational feasibility of classical optimization up to 2048 qubits. Residual full-distribution TVD and sampled MMD^2 errors limit claims of distributional accuracy. More expressive tractable encoders, adaptive kernels and evaluation of quantum deployment are directions for improving conditional generation; classical sampling hardness and practical quantum advantage remain open.

DATA AVAILABILITY

The code and data supporting this study are available in GitHub [48].

ACKNOWLEDGMENTS

OB is a fellow of Eurecat's "Vicente López" PhD grant program.

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Appendix A: Hyperparameters used for the numerical experiments

1. Common training settings

The Gaussian, scaling and profile-selected Curie–Weiss models minimize the unbiased squared-MMD estimator using `Adam` [49], with second-moment coefficient 0.999 and numerical stabilizer 10^{-8}. The first-moment coefficient is 0.9 for the Gaussian and scaling experiments and is tuned for Curie–Weiss. Learning rates follow a cosine decay, and parameters are initialized independently from $\mathcal{N}(0, s_\theta^2)$. Table IV lists the experiment-specific budgets. Section A 3 b describes hyperparameter selection based on full-distribution TVD.

One training run uses an independent seed and includes the prescribed restarts. Each restart resets the optimizer and draws new initial parameters while reusing target pools. Curie–Weiss restarts reuse sampling schedules; scaling restarts use different sampling randomness. We record checkpoints at initialization and the intervals specified below, then select the lowest-scoring checkpoint across restarts. Reported uncertainty is computed across runs, not restarts.

Each update uniformly samples one training condition and one bandwidth, then draws a target minibatch with replacement from its training pool. Nonidentity masks are refreshed at every update, retaining the nonidentity-probability correction. Mask budgets include repetitions. Reference samples are independent of the target minibatch and shared across masks.

TABLE IV. Training and sampled-test budgets per circuit. The Curie–Weiss columns describe profile-TVD selection; Table IX gives the budgets for full-TVD selection. Updates are per restart. Curie–Weiss learning rates, initialization widths and first-moment coefficients are selected by HPO; Gaussian and scaling settings are fixed. Tuned values are rounded to three significant figures. Test budgets refer to sampled MMD diagnostics; small-system TVDs use exact probabilities. Full TVD is the primary reporting metric, while profile TVD retains its role in selection.

Setting	Gaussian	Shared conditional IQP across conditions	Independent conditional IQP per condition	Independent unconditional IQP per condition	Scaling
Updates per restart	300	2,000	2,000	2,000	5,000
Restarts per model and seed	1	3	3	3	3
Final evaluation seeds per model	10	5	5	5	3
Training-pool samples per condition	1,000	10,000	10,000	10,000	10,000
Test target draws per condition/estimate	1,000	2,000	2,000	2,000	2,000
Test MMD estimates per seed and condition grid	1	1	1	1	5
Target minibatch size	100	450	450	450	450
Mask draws per update	1,024	2,048	2,048	2,048	512
Reference draws per update	1,024	2,048	2,048	2,048	1,024
Expected mask weights	$\{1\}$	$\{2, 4, 6\}$	$\{2, 4, 6\}$	$\{2, 4, 6\}$	$\{2, 4\}$
Initial learning rate	5×10^{-3}	0.0401	0.0400	0.0324	0.01
Final learning rate	5×10^{-4}	3.26×10^{-3}	7.83×10^{-4}	3.65×10^{-4}	10^{-3}
Initial phase standard deviation s_θ	0.05	0.430	0.349	0.0758	0.1
First-moment coefficient b_1	0.9	0.968	0.812	0.975	0.9

For expected mask weight w, the bandwidth is

$$\sigma(n, w) = \left[-2 \ln\left(1 - \frac{2w}{n}\right)\right]^{-1/2}, \qquad 0 < w < n/2, \tag{A1}$$

where n counts visible qubits and the expectation precedes conditioning on a nonidentity mask. Bandwidths contribute equally to the objective. All encoders are fixed during training, and rotation angles are in radians. All reported model profiles use raw circuit probabilities. Encoder-only baselines receive no training. Sampled-test target draws are independent of the training pools; these diagnostics do not determine the exact full or profile TVDs.

2. Discrete Gaussian translation

The model has 16 visible qubits on an open 4×4 lattice, with 16 single-qubit and 24 pair phases. Its input is

$$|\phi_{\mathrm{G}}(c)\rangle = \bigotimes_{i=1}^{16} R_Z(c - 1/2) R_Y(1.5) \, |0\rangle_i \, . \tag{A2}$$

The target profile is $\rho(k \mid c) \propto \exp[-(k - (5 + 6c))^2/8]$, with uniform probabilities within each Hamming-weight shell. Training uses $c \in \{0.0, 0.25, 0.50, 0.75, 1.0\}$ and 1,000 samples per condition, giving 5,000 training samples per seed. We select the checkpoint minimizing mean exact profile TVD at development conditions $c \in \{0.125, 0.375, 0.625, 0.875\}$, including initialization and checks every 30 updates. Evaluation uses $c_j = (j + 1/3)/80$ for $j = 0, \ldots, 79$, with 1,000 target draws per condition for auxiliary test-MMD diagnostics. Full and profile TVDs use exact probabilities; each aggregate error is the median across ten independent seeds of the mean error over evaluation conditions. The full-TVD curve shows pointwise medians and interquartile ranges, which need not aggregate to that same value. Checkpoint selection remains based on development-profile TVD. Figure 2 illustrates one run chosen independently of held-out performance.

3. Curie–Weiss comparison

The target law is $p(x \mid \beta) \propto \exp[\beta(2|x| - 16)^2/32]$. All trained comparison models have 16 visible qubits and the same complete set of 16 single-qubit and 120 pair phases. Conditional models use

$$|\phi_{\mathrm{CW}}(\beta)\rangle = \bigotimes_{i=1}^{16} R_Z(0.75\beta) R_Y(\pi/2) \, |0\rangle_i \, . \tag{A3}$$

The shared conditional IQP across conditions trains at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$ and is evaluated at $\beta \in \{0.4, 1.0, 1.6\}$.

Independent baselines fit a separate circuit at each reporting temperature, using either the corresponding conditional input or $|0\rangle^{\otimes 16}$. Each circuit has 10,000 training samples and uses 2,000 updates per restart with three restarts in the final evaluation. Each baseline family therefore requires three circuits per seed. Table V distinguishes these aggregate budgets. Independent models are evaluated at fitted temperatures, whereas the shared conditional IQP across conditions is evaluated at temperatures omitted from fitting and selection. Auxiliary test-MMD diagnostics use 2,000 independently drawn targets per reported temperature. Full and profile TVDs use exact probabilities; final results report means and sample standard deviations across five independent seeds. Full TVD is the primary reported accuracy metric, while HPO and checkpoint selection retain profile TVD.

TABLE V. Curie–Weiss comparison budgets per seed, summed over fitting conditions. Training samples count target-pool entries, rather than repeated minibatch draws. Optimizer updates include all three restarts of every trained circuit.

Model	Trained circuits	Training samples	Optimizer updates
Encoder only	0	0	0
Shared conditional IQP across conditions	1	40,000	6,000
Independent conditional IQP per condition	3	30,000	18,000
Independent unconditional IQP per condition	3	30,000	18,000

a. Hyperparameter optimization based on profile TVD

We use Optuna's tree-structured Parzen estimator [50] to tune cosine-decay `Adam` and the Gaussian initialization width, while fixing the encoder, the 136-parameter ansatz, the training pools, kernel weights, and sampling budgets. The shared-model search uses 48 trials, each completing 2,000 sampled-MMD updates on each of two screening seeds, with one restart and no early pruning. The score averages the best exact development-profile TVD across the two seeds. Checkpoint selection includes initialization and every 25 updates at development temperatures $\{0.5, 0.7, 1.3, 1.5\}$. We also tried `AdamWScheduleFree` [51] with the same HPO trial budget, but it did not outperform cosine-decay `Adam` on the development-profile TVD.

TABLE VI. `Adam` HPO search ranges for the Curie–Weiss models. Learning rates, initialization widths, and decay ratios are sampled logarithmically; b_1 is sampled uniformly.

Setting	Search range
Initial learning rate	$[3 \times 10^{-4}, 5 \times 10^{-2}]$
Gaussian initialization width s_θ	$[0.01, 0.7]$
First-moment coefficient b_1	$[0.8, 0.98]$
Final/initial learning-rate ratio	$[0.01, 0.3]$

Table VI gives the `Adam` search ranges. Each study uses 12 startup trials and batches of two proposals, with sampler updates ordered independently of completion order. The second-moment coefficient, numerical stabilizer, and weight decay are fixed at 0.999, 10^{-8}, and zero. We confirm the two best distinct `Adam` configurations on three fresh seeds, with three restarts per seed, and select the lowest mean development-profile TVD. The selected settings in Table IV are then fixed for evaluation on five further independent seeds, with three restarts per seed and development-based checkpoint selection. Reporting temperatures $\{0.4, 1.0, 1.6\}$ do not enter this search or selection, although they had been examined in earlier experiments.

For each independent baseline family, we run one `Adam` HPO study with 48 trials and the same search ranges and per-circuit budgets. Each trial fits circuits at $\beta = 0.4, 1.0, 1.6$ on two screening seeds and scores the mean best fitted-temperature profile TVD across these six fits. One configuration therefore serves all temperatures within a family. Each baseline study includes a fixed `Adam` reference configuration and the selected shared-model configuration within its trial budget. The shared-model study includes the same fixed reference configuration. We confirm the two best configurations per family on three fresh seeds with three restarts per circuit, select the lowest mean fitted-temperature profile TVD, then evaluate on five further independent seeds with three restarts per circuit. Checkpoint selection includes initialization and every 25 updates at the fitted temperature. Equal trial counts require three times as many circuit fits per independent family as for the shared-model `Adam` search. Reporting temperatures remain in-sample; fresh seeds measure optimization variability.

At $\beta = 1.6$, exact full TVDs across the five final runs range over 0.1671–0.4012 for the shared model, 0.4524–0.5388 for independent conditional IQPs, and 0.3439–0.4027 for independent unconditional IQPs. The shared model has lower full TVD than the independent conditional model in all five paired runs and than the independent unconditional model in four of five. At $\beta = 1.0$, it has lower full TVD than both independent families in all five paired runs. These comparisons describe optimization variability under the existing profile-based selection protocol.

TABLE VII. Exact raw Curie–Weiss hamming-profile TVD at sixteen qubits. Trained rows report means and sample standard deviations over 5 independent runs; the encoder is deterministic. Shared reporting temperatures are held out, while independent circuits are fitted at each reporting temperature. Bold entries mark the lowest mean error, including the encoder. Selection remains profile TVD.

Model	Evaluation	Hamming-profile TVD ↓		
		$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Encoder only	Baseline	0.1180	0.4406	0.8699
Shared conditional IQP across conditions	Held-out	0.0712 ± 0.0158	0.0683 ± 0.0153	$\mathbf{0.1530 \pm 0.1245}$
Independent conditional IQP per condition	Fitted	0.0530 ± 0.0169	0.0334 ± 0.0088	0.2957 ± 0.0377
Independent unconditional IQP per condition	Fitted	$\mathbf{0.0170 \pm 0.0114}$	$\mathbf{0.0296 \pm 0.0080}$	0.3500 ± 0.0094

Table VII retains the secondary exact profile errors. At $\beta = 0.4, 1.0, 1.6$, the shared model gives 0.0712 ± 0.0158, 0.0683 ± 0.0153, and 0.1530 ± 0.1245. Both independent families have lower mean profile error at $\beta = 0.4$ and 1.0, whereas the shared model has lower mean profile error at $\beta = 1.6$. These rankings differ from the primary full-TVD rankings at $\beta = 1.0$. All spreads are sample standard deviations across the same five seeds. Full TVD was evaluated without retuning or changing checkpoint selection.

b. Full-distribution TVD hyperparameter optimization

For hyperparameter selection based on exact full-distribution TVD, we use the following protocol. The results in Tables II and III use the profile-based selection described above and in Sec. A 3 c.

We run four separate studies: a shared model with the product phase input, a shared model with its normalized +1 global spin-flip projection, an independent unconditional baseline family, and an independent conditional baseline family. The projected input is the spin-flip construction in Sec. IV B 1. All circuits use 16 visible qubits, no ancillas, and the same 136 independently parameterized one- and two-body phases. The shared models fit at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$ and select on development temperatures $\{0.5, 0.7, 1.3, 1.5\}$. Each independent family fits three circuits at $\beta = 0.4, 1.0, 1.6$, selecting each circuit at its fitted temperature and sharing one hyperparameter configuration across the family.

Each study uses 60 Optuna TPE trials with a categorical optimizer choice among `Adam`, `AdamWScheduleFree`, and `SGDScheduleFree` [49–51].

We allocate the same trial budget to all four studies, searching only three hyperparameters: the optimizer, learning rate and Gaussian initialization width. As a random-search benchmark [52], 60 independent draws have probability $1 - (1 - 0.05)^{60} \approx 0.954$ of sampling at least one configuration in the best-performing 5% of the chosen sampling distribution. This benchmark motivates the budget; it is not a coverage guarantee for our adaptive TPE proposals, which use previous scores to prioritize promising configurations.

Table VIII gives the search space. Each fit uses a constant learning rate and independent initial phases $\theta_j \sim \mathcal{N}(0, s_\theta^2)$. We fix $b_1 = 0.9$, $b_2 = 0.999$ and $\epsilon = 10^{-8}$ where applicable, zero weight decay, no warmup, and no gradient clipping. The schedule-free optimizers use learning-rate weighting exponent 2 and their evaluation parameters for checkpoint scoring.

TABLE VIII. Search space for Curie–Weiss hyperparameter optimization based on full-distribution TVD. The optimizer is categorical; learning rates and initialization widths are sampled logarithmically. The 60 trials per study have no allocation quota by optimizer.

Hyperparameter	Options or range
Optimizer	`Adam`, `AdamWScheduleFree`, `SGDScheduleFree`
Learning rate, `Adam` or `AdamWScheduleFree`	$[3 \times 10^{-4}, 10^{-1}]$
Learning rate, `SGDScheduleFree`	$[10^{-3}, 1]$
Gaussian initialization width s_θ	$[0.01, 0.7]$ radians

Each regular trial runs 1000 updates on three seeds, with one initialization per seed. We evaluate exact TVD over all 2^{16} bitstrings at initialization and every 100 updates. For a shared model, each seed selects the checkpoint with the lowest mean development-temperature TVD. For an independent family, each circuit selects its lowest fitted-temperature TVD, then the three fitted-temperature errors are averaged within each seed. The trial score is the median of these three seed scores. TPE uses 12 startup trials, including one reference configuration per optimizer, and batches of two proposals with sampler updates in proposal order. No trial is pruned. The reference learning rates are 0.01 for the `Adam` variants and 0.1 for `SGDScheduleFree`, all with $s_\theta = 0.5$.

We promote the three best distinct optimizer, learning-rate and initialization-width configurations to confirmation on five fresh seeds. Each confirmation fit uses 2000 updates and three total initializations per seed, selecting the best checkpoint across those initializations. The winner minimizes the median of the five seed scores. Reporting evaluates the winning configuration's selected confirmation checkpoints at $\beta = 0.4, 1.0, 1.6$, without further fitting or reselection. Thus, the five reporting seeds are reused from configuration selection rather than forming an independent final cohort. Shared reporting temperatures remain excluded from fitting and selection, although they have historical exposure. Independent reporting temperatures remain in-sample. Per-temperature results report means and sample standard deviations across the five seeds; restarts and temperatures are not additional seed replicates.

TABLE IX. Sampling budgets for Curie–Weiss hyperparameter optimization based on full-distribution TVD. The same budgets apply to all four studies during screening and confirmation. Target-pool sizes are per condition and seed; minibatches are drawn with replacement.

Setting	Budget
Training target pool per fitting condition	1024
Target minibatch per update	256
Mask draws per update	2048
Reference draws per update	2048
Validation target pool per fitting condition	2048
Test targets per reporting condition	2048
Diagnostic masks per condition and bandwidth	2048
Diagnostic reference draws per condition	2048

Training minimizes the unbiased squared-MMD objective using the budgets in Table IX. Each update uniformly samples one fitting temperature and one bandwidth. Masks are refreshed at every update and retain the nonidentity-probability correction. Reference draws are shared across masks and independent of target minibatches. The actual expected mask weights are $\{2, 4, 7.99\}$ before nonidentity conditioning; the displayed label 8 rounds 7.99. All three components therefore use strictly positive Gaussian bandwidths.

Auxiliary sampled-MMD diagnostics use 2048 validation targets per fitting temperature and independent test targets of the same size per reporting temperature, 2048 masks per condition and bandwidth, and 2048 reference draws per condition shared across bandwidths. These diagnostics do not select checkpoints or configurations. Parameter and optimizer states use double precision; phase contractions use single precision with sample chunks of 256.

At equal trial counts, each independent family requires three times as many circuit fits as a shared model.

c. *Matched comparison of product and spin-flip invariant inputs*

For deterministic preparation, write $c = \cos(t/2)$, $s = \sin(t/2)$, and let $Z_\ell^{(p)} = [1 + (-1)^p \cos^\ell(t)]/2$ be the parity-p probability of ℓ independent reference bits. The last register wire accumulates the parity p of the prepared prefix. With ℓ wires remaining, the next wire receives a parity-controlled $R_Y(\alpha_p)$, where

$$\alpha_p = 2\,\mathrm{atan2}\left(s\sqrt{Z_{\ell-1}^{(p\oplus 1)}}, c\sqrt{Z_{\ell-1}^{(p)}}\right). \tag{A4}$$

A CNOT from that wire updates the accumulator. Repeating on the first $n-1$ wires leaves the last wire equal to their parity and prepares the even-conditioned reference magnitudes. Applying a phase of $-\pi/2$ to each occupied wire gives Eq. (50); final Hadamards prepare Eq. (49). This construction uses $O(n)$ one- and two-qubit gates within the original register.

The spin-flip comparison uses the two inputs in Sec. IV B 1, $n = 16$ visible qubits, no ancillas, and the same complete 136-parameter ansatz. Each input has its own 48-trial cosine-`Adam` search over the ranges in Table VI. Both studies include a fixed `Adam` reference configuration and the shared-model configuration in Table IV within that budget. Optuna TPE uses 12 startup trials, one independent sampler seed per input, two proposals per batch, no pruning, and sampler updates in proposal order.

Screening uses two seeds and one 2000-update restart per candidate and seed. We confirm the two best distinct configurations for each input on three fresh confirmation seeds, with three restarts per seed. The selected configuration minimizes mean exact development-profile TVD. We fix both configurations before final evaluation on five fresh paired evaluation seeds, with three restarts per seed. Selection includes initialization and every 25 updates at development temperatures $\{0.5, 0.7, 1.3, 1.5\}$; fitting uses $\{0.6, 0.8, 1.2, 1.4\}$. Reporting temperatures $\{0.4, 1.0, 1.6\}$ enter neither fitting nor current selection but have historical exposure. The completed comparison used 258 restart trajectories and 516,000 optimizer updates. Table X gives the selected hyperparameters and their confirmation scores.

TABLE X. Selected `Adam` hyperparameters. Scores are mean exact development-profile TVDs on confirmation seeds. Settings are rounded to three significant figures.

Input	Initial LR	s_θ	b_1	Final/initial LR	Selection score
Product-state input	0.0456	0.525	0.964	0.279	0.0524
Spin-flip invariant input	0.0485	0.633	0.98	0.0394	0.0349

Each arm retains 10,000 target-pool entries per fitting temperature, batch size 450, 2048 nonidentity-mask draws and 2048 reference draws per update, and expected mask weights $\{2, 4, 6\}$. Paired seeds share target pools, minibatch, condition, bandwidth and mask schedules, and Gaussian initialization directions; selected initialization widths may differ. Reference samples use common random numbers with input-dependent sampling laws. We use the Hamming-RBF mask law, including odd masks whose projected-model moments vanish. Sampled validation diagnostics retain 2000 target samples, 256 masks and 16,384 reference draws; development-profile TVD determines selection.

As a comparator without additional training, we average the product-input model's output with its complement, $q_{\mathrm{avg}}(x \mid \beta) = [q(x \mid \beta) + q(\bar{x} \mid \beta)]/2$, without retraining or reselection. This law can be sampled by complementing each model output with probability 1/2. It satisfies spin-flip symmetry, whereas the invariant encoder imposes that symmetry on the pure input state before learning. Both deterministic encoder-only laws are also evaluated. Final metrics use exact raw probabilities, prioritize full-distribution TVD, and report mean and sample standard deviation over the five paired seeds; seed-wise differences concern optimization variability within this specified temperature family.

To distinguish the two operations, let $A_{\boldsymbol{\theta},\beta}(x) = \langle x|U_{\boldsymbol{\theta}}|\phi(\beta)\rangle$ be the product-input output amplitude. At the same circuit phases, the invariant input gives

$$q_{\boldsymbol{\theta},+}(x \mid \beta) = \frac{|A_{\boldsymbol{\theta},\beta}(x) + A_{\boldsymbol{\theta},\beta}(\bar{x})|^2}{2[1+\cos^n(t)]}. \tag{A5}$$

The interference term changes the model family relative to classical probability averaging. For the spin-flip invariant target, convexity guarantees that output averaging cannot increase either profile or full-bitstring TVD of the product-input model. Input projection has no corresponding guarantee of improved fit.

The product encoder alone gives the uniform law. The invariant encoder alone gives

$$q_{0,+}(x \mid \beta) = \frac{1+\cos[(2|x|-n)t]}{2^n[1+\cos^n(t)]}. \tag{A6}$$

Both encoder-only laws are uniform within Hamming shells and spin-flip invariant. Their profile and full-bitstring TVDs therefore coincide. At $\beta = 0.4, 1.0, 1.6$, these errors are 0.1180, 0.4406, and 0.8699 for the product input and 0.2561, 0.5536, and 0.9002 for the invariant input. Projection enforces the physical symmetry but does not improve these untrained baselines.

TABLE XI. Exact encoder-only errors and diagnostics after averaging the trained product-input output with its complement. Full TVD is primary and Profile TVD is secondary. Encoder-only rows are deterministic. Averaged outputs use the same checkpoints without additional fitting or selection. Means ± sample standard deviations over 5 paired seeds. Reporting temperatures enter neither fitting nor current selection but have historical exposure.

Model	Metric	$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Product-state input, encoder only	Full TVD	0.1180	0.4406	0.8699
	Profile TVD	0.1180	0.4406	0.8699
	D_{flip}	0.0000	0.0000	0.0000
Spin-flip invariant input, encoder only	Full TVD	0.2561	0.5536	0.9002
	Profile TVD	0.2561	0.5536	0.9002
	D_{flip}	0.0000	0.0000	0.0000
Product output, classically averaged	Full TVD	0.0585 ± 0.0066	0.2010 ± 0.0164	0.1921 ± 0.0217
	Profile TVD	0.0223 ± 0.0048	0.0645 ± 0.0112	0.0986 ± 0.0154
	D_{flip}	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000

Table XI reports the encoder-only and output-averaging diagnostics. At $\beta = 0.4$, averaging the trained product-input output gives full TVD 0.0585 ± 0.0066 and profile TVD 0.0223 ± 0.0048, both lower than for the invariant input. At $\beta = 1.0$ and 1.6, its full TVDs are 0.2010 ± 0.0164 and 0.1921 ± 0.0217, respectively, compared with 0.1645 ± 0.0140 and 0.1746 ± 0.0182 for the invariant input. The corresponding profile errors remain close. These comparisons separate coherent input projection from classical output averaging under the specified profile-based selection protocol.

Table XII reports paired differences, invariant input minus product input. Full TVD improves in all five paired seeds at $\beta = 0.4$ and 1.0, and in four of five at $\beta = 1.6$. Profile differences change sign across seeds at every reporting temperature. Sample standard deviations describe the paired differences rather than differences of the separate standard deviations.

TABLE XII. Paired exact-error differences, spin-flip invariant input minus product-state input. Profile and Full denote profile and full-bitstring TVD; Full TVD is primary and Profile TVD is secondary. Negative differences favor the invariant input. Mean and sample SD summarize the paired differences across 5 paired seeds. Reporting temperatures enter neither fitting nor current selection but have historical exposure.

Statistic	$\beta = 0.4$			$\beta = 1.0$			$\beta = 1.6$		
	Full	Profile	D_{flip}	Full	Profile	D_{flip}	Full	Profile	D_{flip}
Mean	−0.1450	−0.0340	−0.4837	−0.0547	0.0014	−0.1586	−0.0190	−0.0025	−0.0183
Sample SD	0.0097	0.0271	0.0133	0.0201	0.0195	0.0184	0.0345	0.0239	0.0017

4. Scaling

The scaling models use the Curie–Weiss encoder on every wire, with no ancillas and open lattices listed in Table XIII. At size n, the target law is $p(x \mid \beta) \propto \exp[\beta(2|x|-n)^2/(2n)]$. Training uses $\beta \in \{0.6, 0.8, 1.2, 1.4\}$, with 10,000 samples per condition and 40,000 per seed. Evaluation uses $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$. We use three independent seeds at each size. Table IV gives the cosine-`Adam` settings, with 5,000 updates per restart and three restarts per seed. These settings are retained independently of the sixteen-qubit HPO; their transfer or retuning has not been evaluated.

TABLE XIII. Scaling architectures, including single-qubit and nearest-neighbor pair phases.

Visible qubits	Lattice	Phases
64	8×8	176
128	8×16	360
256	16×16	736
512	16×32	1,488
1,024	32×32	3,008
2,048	32×64	6,048

Checkpoint selection minimizes mean sampled validation MMD^2 across training temperatures and bandwidths, checking initialization and every 25 updates across three restarts. Held-out temperatures do not enter fitting or selection. We report the selected restart at initialization and every 500 updates through update 5,000, averaging five independent MMD estimates per seed and condition grid. Each estimate uses 2,000 target draws per condition, 512 masks per condition and bandwidth, and 1,024 reference draws per condition. The reporting sample sets are reused across iterations; reference samples are shared across bandwidths. The plot shows the median and interquartile range across three seed averages. We retain negative unbiased MMD^2 estimates.